

Dynamic Contextual Pricing Engine with Multi-Armed Bandits (Thompson Sampling)

Abstract

This report presents the implementation of a **Dynamic Contextual Pricing Engine** using the **Multi-Armed Bandit** approach with **Thompson Sampling**. The system dynamically selects optimal product prices by balancing exploration and exploitation, maximizing expected revenue based on customer responses while continuously learning from historical interactions.

Introduction

Dynamic pricing is a strategy in which product prices are adjusted based on customer behavior, demand, and market conditions. Traditional pricing methods rely on fixed rules, whereas Multi-Armed Bandit algorithms learn optimal pricing decisions over time. This project implements a contextual pricing engine using Thompson Sampling to improve pricing decisions and revenue generation.

Theory

The Multi-Armed Bandit problem models pricing decisions as selecting one option (price) from multiple alternatives. **Thompson Sampling** is a Bayesian algorithm that:

- Models uncertainty using probability distributions.
- Balances exploration of new prices and exploitation of profitable prices.
- Updates beliefs based on customer purchase outcomes.
- Learns the optimal pricing strategy over time.

Objectives

- Implement a Dynamic Contextual Pricing Engine using Thompson Sampling.
- Model multiple pricing options as bandit arms.
- Learn customer purchase behavior from observed rewards.
- Maximize expected revenue through adaptive pricing.
- Evaluate pricing performance using simulated customer interactions.

Methodology

1. Define a set of possible product prices as bandit arms.
2. Initialize Bayesian prior distributions for each pricing option.
3. Observe customer context and select a price using Thompson Sampling.
4. Record purchase outcomes and revenue as rewards.

5. Update posterior distributions after each interaction.
6. Continuously improve pricing decisions through repeated learning.

Dataset

A synthetic customer interaction dataset is used to simulate purchase behavior under different pricing options. Each record contains customer context, selected price, purchase outcome, and reward. The same framework can be extended to real-world e-commerce or retail transaction datasets.

Algorithm

The Thompson Sampling algorithm performs the following steps:

- Sample a probability value from the posterior distribution of each pricing arm.
- Select the arm with the highest sampled value.
- Observe the customer response and calculate the reward.
- Update the corresponding posterior distribution based on the observed outcome.
- Repeat the process to continuously optimize pricing decisions.

Complexity Analysis

For K pricing options, Thompson Sampling requires $O(K)$ time for price selection at each interaction. Posterior updates are performed in constant time, making the algorithm computationally efficient and scalable for real-time pricing applications.

Results

The implemented pricing engine successfully learns customer preferences, adapts pricing strategies over time, balances exploration and exploitation, increases cumulative rewards, and identifies pricing options that maximize expected revenue.

Conclusion

The Dynamic Contextual Pricing Engine demonstrates the effectiveness of Thompson Sampling for adaptive pricing. Its Bayesian learning framework enables continuous optimization of pricing decisions, making it suitable for e-commerce, retail, subscription services, and other revenue management applications.

References

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